جامعة أم القرى
UMM AL-QURA UNIVERSITY

Computing Club

Introductionto DatabaseSecond Semester
2026/1447

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Group 7

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Task Schedule

Name	Tasks
Ghala	• Conceptual Design • Member Table
Jana	• ER Model • Community Table
Sara	• File Design • Activity Table + Resource
Jawana	• Relational Schema Diagram • Relationship Table

Section 1

Conceptual Design

System description:

The computing club database system is designed to manage the main data and activities of the computing club. It stores information about members, the club community. And club activities. It also keeps track of resources used for each Activity. The system supports organizing activities. Recording member participation, and linking resources to activities. **Required identification:**

1. **Entities** :
   - Member
   - Community
   - Activity
   - Resource
   - Member activity

2. **Attributes**
   - **Member:** Academic Id, Fname, Lname, Email. Phone, Role, College, Major
   - **Community:** Community_Name. Description, Contact Email
   - **Activity:** Activity Id. Activity Name, Description, Date, Time, Location, Type
   - **Resource:** Resource Id, Resource Name, Resource Type, Supplier Name

3. **Relationships**

1. community organizes activity (1:N)
   - One community can organize many activities.
   - Each activity belongs to one community

2. Member participates in activity (M:N)
   - A member can participate in many activities
   - This is presented in the relational scheme using member member_activity

3. Activity uses resources (1:N)
   - One activity can use many resources
   - Each resources is linked to one activity

ER MODEL

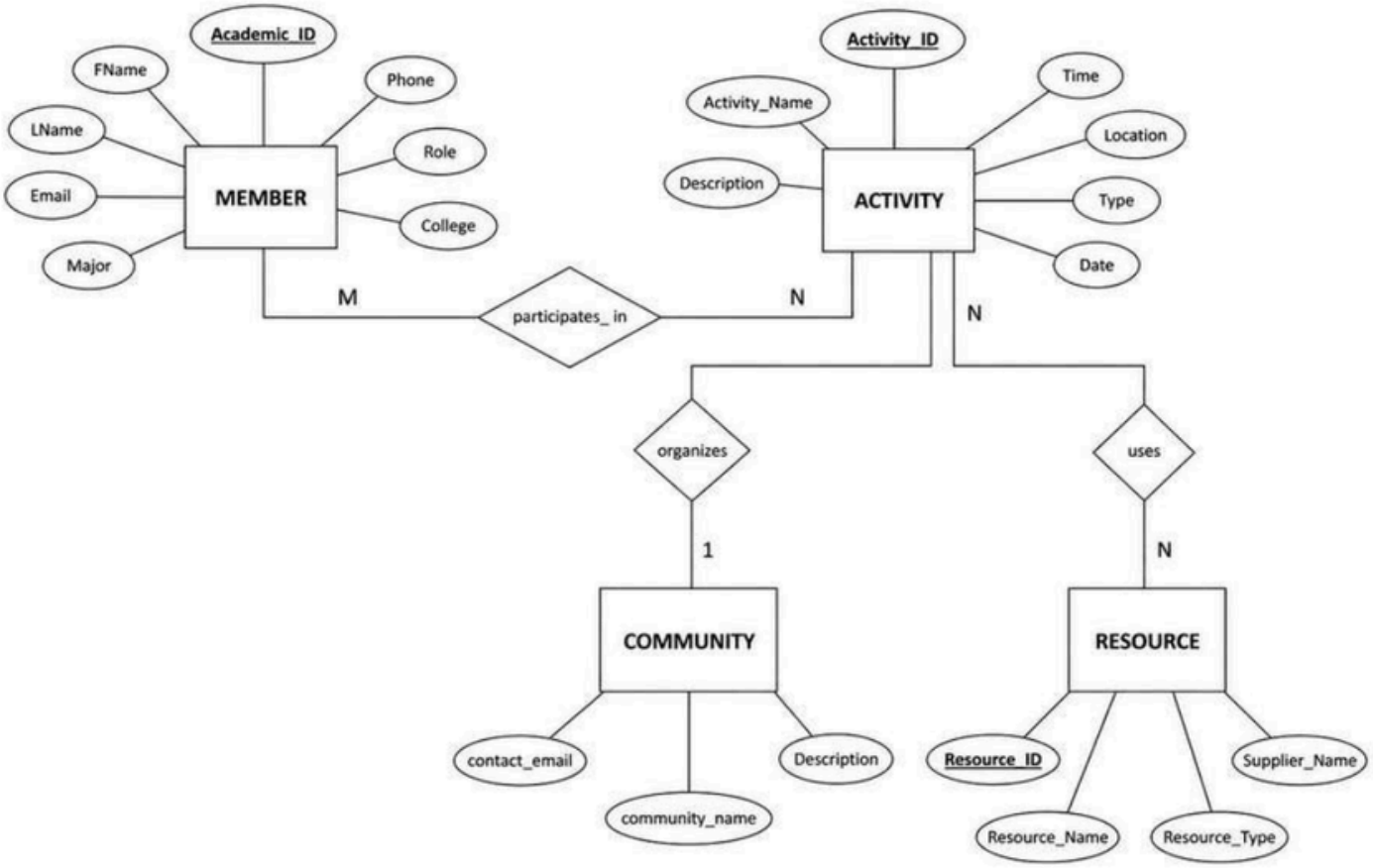


Image 1: Computing Club Database ER Model

Relational Schema Diagram

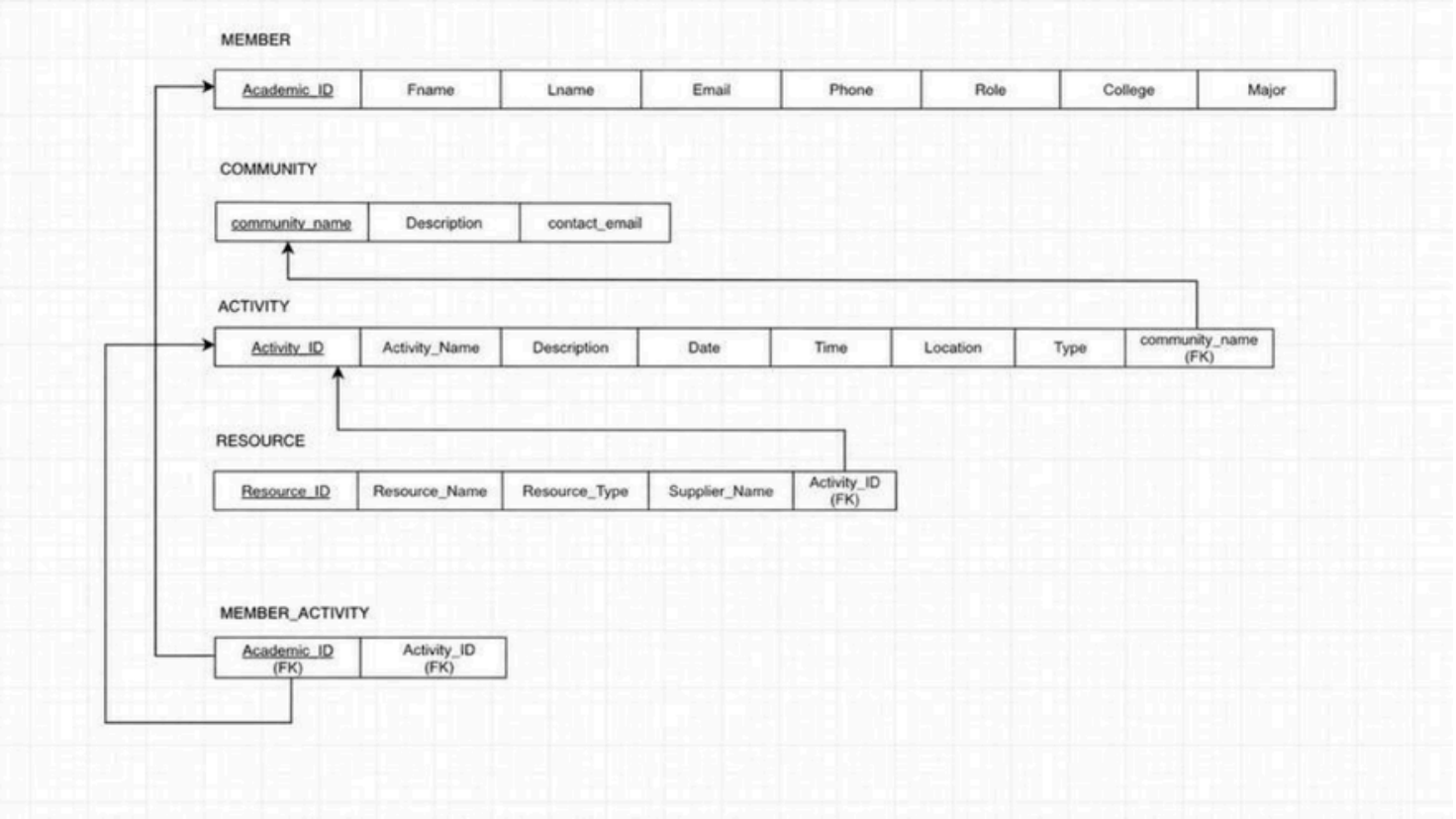


Image 2: Relational Schema Diagram for Computing Club Management System

Section 2

Screenshots of SQL commands and their results:

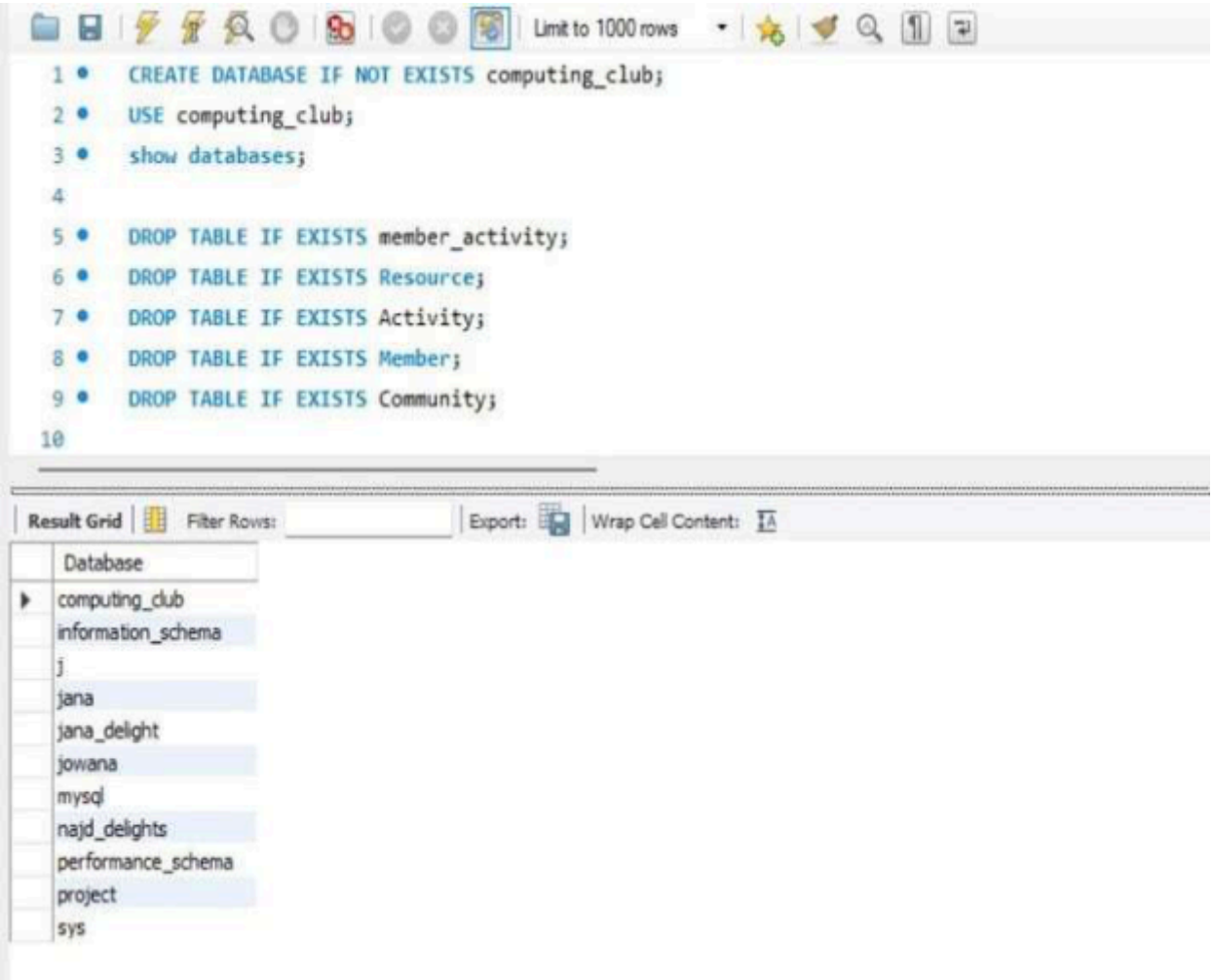


Image 3: Database Creation and Initialization

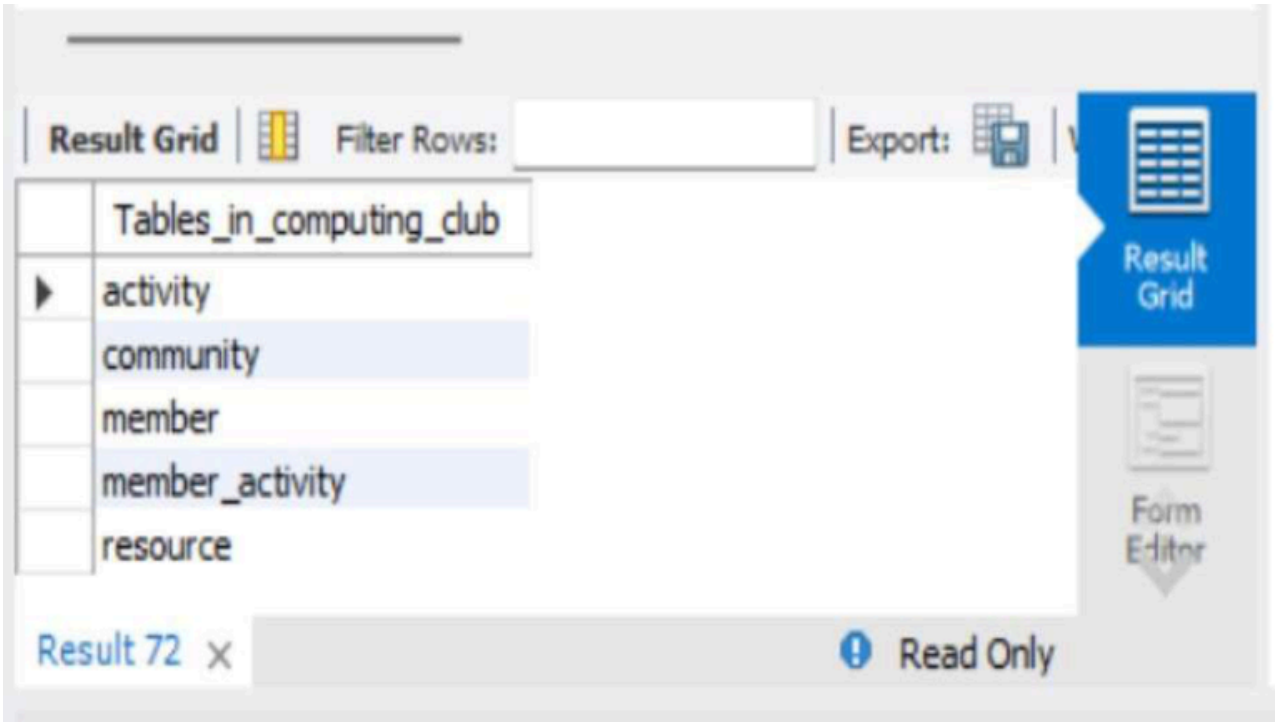


Image 4: Database Tables List

Describe Member

```
13  -- ======================================
14  CREATE TABLE Member (
15  Academic_ID INT PRIMARY KEY,
16  Fname VARCHAR(50) NOT NULL,
17  Lname VARCHAR(50) NOT NULL,
18  Email VARCHAR(100) NOT NULL UNIQUE,
19  Phone VARCHAR(15) CHECK (Phone LIKE '05%'),
20  Role VARCHAR(50) NOT NULL CHECK (Role IN ('Leader', 'Member', 'Organizer', 'Volunteer')),
21  College VARCHAR(100) NOT NULL,
22  Major VARCHAR(100) NOT NULL
23  ) ENGINE=InnoDB;
24
25  describe Member;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Academic_ID	int	NO	PRI	NULL	
Fname	varchar(50)	NO		NULL	
Lname	varchar(50)	NO		NULL	
Email	varchar(100)	NO	UNI	NULL	
Phone	varchar(15)	YES		NULL	
Role	varchar(50)	NO		NULL	
College	varchar(100)	NO		NULL	
Major	varchar(100)	NO		NULL	

Image 5: Describe Member

Select from Member

```
25  describe Member;
26
27  INSERT INTO Member VALUES
28  (4461001, 'Sara', 'Ali', 'sara1@uqu.edu.sa', '0501111111', 'Leader', 'Computer Science', 'Database'),
29  (4461002, 'Jana', 'Ahmed', 'jana2@uqu.edu.sa', '0502222222', 'Member', 'Computer Science', 'Artificial Intelligence'),
30  (4461003, 'Jawana', 'Saad', 'jawana3@uqu.edu.sa', '0503333333', 'Organizer', 'Computer Science', 'Cybersecurity'),
31  (4461004, 'Ghala', 'Hussein', 'ghala4@uqu.edu.sa', '0504444444', 'Member', 'Computer Science', 'Software Engineering'),
32  (4461005, 'Layan', 'Omar', 'layan5@uqu.edu.sa', '0505555555', 'Volunteer', 'Computer Science', 'Information Systems');
33  select * from Member;
34  -- ======================================
35  -- 2. Community
36  -- ======================================
```

Result Grid

Academic_ID	Fname	Lname	Email	Phone	Role	College	Major
4461001	Sara	Ali	sara1@uqu.edu.sa	0501111111	Leader	Computer Science	Database
4461002	Jana	Ahmed	jana2@uqu.edu.sa	0502222222	Member	Computer Science	Artificial Intelligence
4461003	Jawana	Saad	jawana3@uqu.edu.sa	0503333333	Organizer	Computer Science	Cybersecurity
4461004	Ghala	Hussein	ghala4@uqu.edu.sa	0504444444	Member	Computer Science	Software Engineering
4461005	Layan	Omar	layan5@uqu.edu.sa	0505555555	Volunteer	Computer Science	Information Systems

Image 6: Select from Member

Describe Community

```
-- ----------------------------------------
-- 2. Community
-- ----------------------------------------
CREATE TABLE Community (
Community_Name VARCHAR(100) PRIMARY KEY,
Description VARCHAR(255) NOT NULL,
Contact_Email VARCHAR(100) NOT NULL UNIQUE
) ENGINE=InnoDB;

describe Community;

INSERT INTO Community VALUES
```

Field	Type	Null	Key	Default	Extra
Community_Name	varchar(100)	NO	PRI		
Description	varchar(255)	NO			
Contact_Email	varchar(100)	NO	UNI		

Image 7: Describe Community

Select from Community

```
INSERT INTO Community VALUES
('Programming Club', 'Focuses on coding and software development', 'programming@uqu.edu.sa'),
('Design Club', 'Interested in UI/UX and graphic design', 'design@uqu.edu.sa'),
('Robotics Club', 'Works on robotics and AI projects', 'robotics@uqu.edu.sa'),
('Media Club', 'Handles media, photography and editing', 'media@uqu.edu.sa'),
('Volunteer Club', 'Organizes charity and volunteer events', 'volunteer@uqu.edu.sa');
select * from Community;
```

Community_Name	Description	Contact_Email
Design Club	Interested in UI/UX and graphic design	design@uqu.edu.sa
Media Club	Handles media, photography and editing	media@uqu.edu.sa
Programming Club	Focuses on coding and software development	programming@uqu.edu.sa
Robotics Club	Works on robotics and AI projects	robotics@uqu.edu.sa
Volunteer Club	Organizes charity and volunteer events	volunteer@uqu.edu.sa

Image 8: Select from Community

Describe Activity

```
-- 3. Activity
-- ------------------------------------
CREATE TABLE Activity (
Activity_ID INT PRIMARY KEY,
Activity_Name VARCHAR(100) NOT NULL,
Description VARCHAR(255),
Date DATE NOT NULL,
Time TIME NOT NULL,
Location VARCHAR(100) NOT NULL,
Type VARCHAR(50) CHECK (Type IN ('Workshop', 'Training', 'Event')),
Community_Name VARCHAR(100) NOT NULL,
FOREIGN KEY (Community_Name) REFERENCES Community(Community_Name)
ON DELETE CASCADE
ON UPDATE CASCADE
) ENGINE=InnoDB;
describe Activity;
```

Field	Type	Null	Key	Default	Extra
Activity_ID	int	NO	PRI		
Activity_Name	varchar(100)	NO			
Description	varchar(255)	YES			
Date	date	NO			
Time	time	NO			
Location	varchar(100)	NO			
Type	varchar(50)	YES			
Community_Name	varchar(100)	NO	MUL		

Image 9: Describe Activity

Select from Activity

```
describe Activity;

INSERT INTO Activity VALUES
(1, 'Coding Workshop', 'Learn coding basics', '2026-04-20', '10:00:00', 'Lab 1', 'Workshop', 'Programming Club'),
(2, 'UI Design', 'Design basics', '2026-04-21', '12:00:00', 'Lab 2', 'Workshop', 'Design Club'),
(3, 'Robot Build', 'Build robots', '2026-04-22', '02:00:00', 'Lab 3', 'Training', 'Robotics Club'),
(4, 'Video Editing', 'Edit videos', '2026-04-23', '11:00:00', 'Lab 4', 'Workshop', 'Media Club'),
(5, 'Charity Event', 'Help community', '2026-04-24', '01:00:00', 'Hall A', 'Event', 'Volunteer Club');
select * from Activity;
-- ------------------------------------
```

Activity_ID	Activity_Name	Description	Date	Time	Location	Type	Community_Name
1	Coding Workshop	Learn coding basics	2026-04-20	10:00:00	Lab 1	Workshop	Programming Club
2	UI Design	Design basics	2026-04-21	12:00:00	Lab 2	Workshop	Design Club
3	Robot Build	Build robots	2026-04-22	02:00:00	Lab 3	Training	Robotics Club
4	Video Editing	Edit videos	2026-04-23	11:00:00	Lab 4	Workshop	Media Club
5	Charity Event	Help community	2026-04-24	01:00:00	Hall A	Event	Volunteer Club

Image 10: Select from Activity

Describe Resource

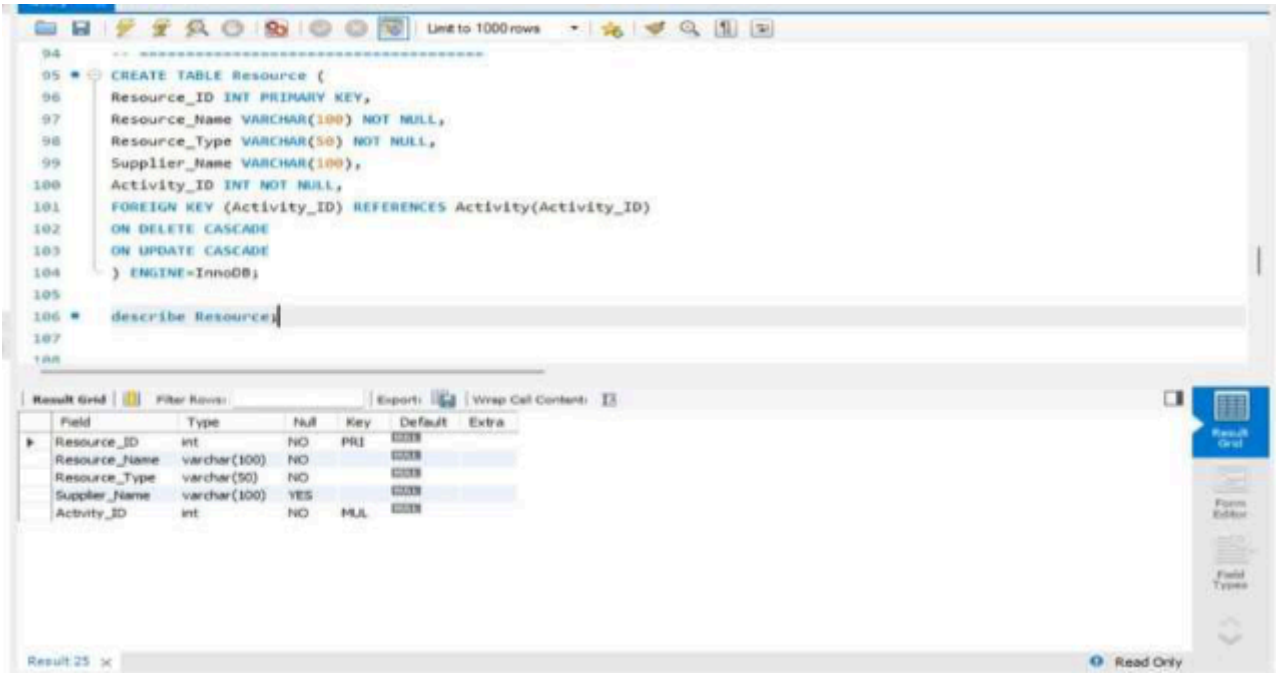


Image 11: Describe Resource

Select from Resource

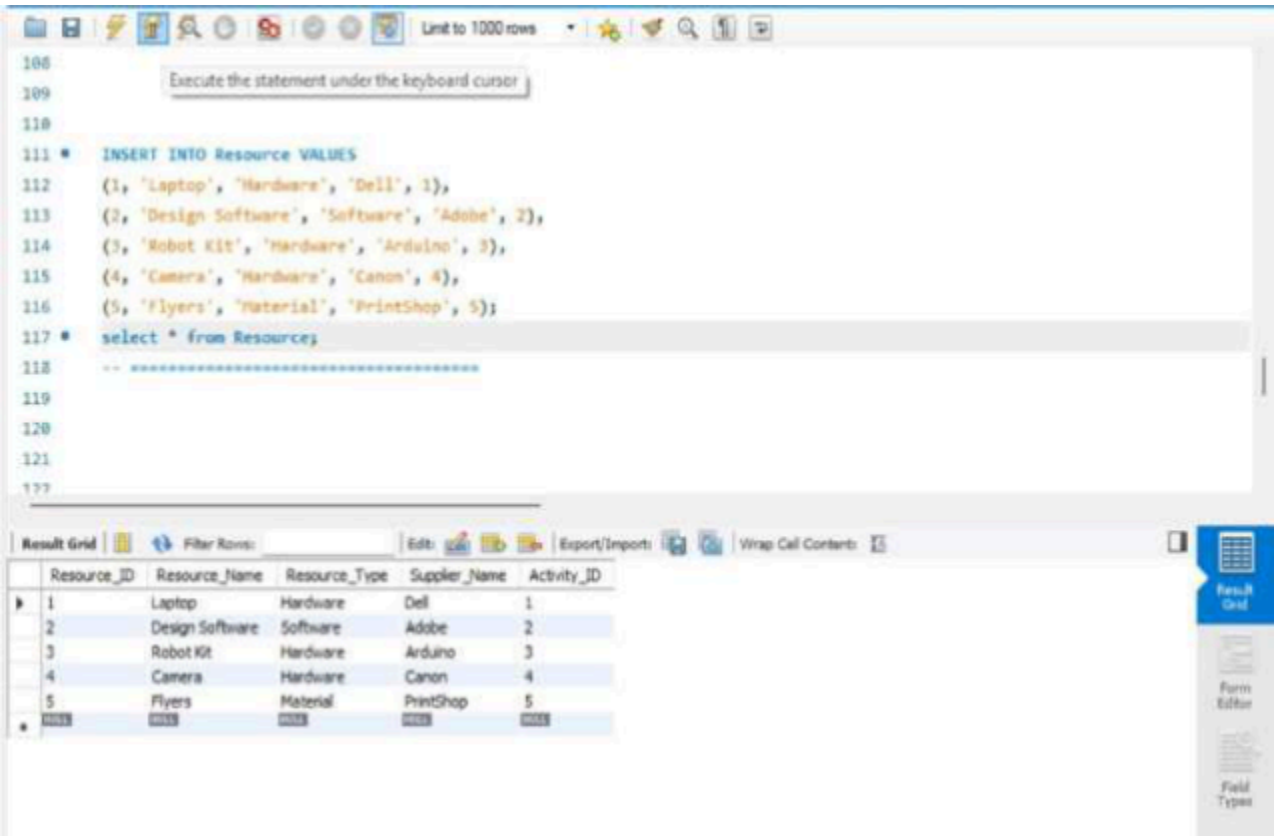


Image 12: Select from Resource

Describe member_activity

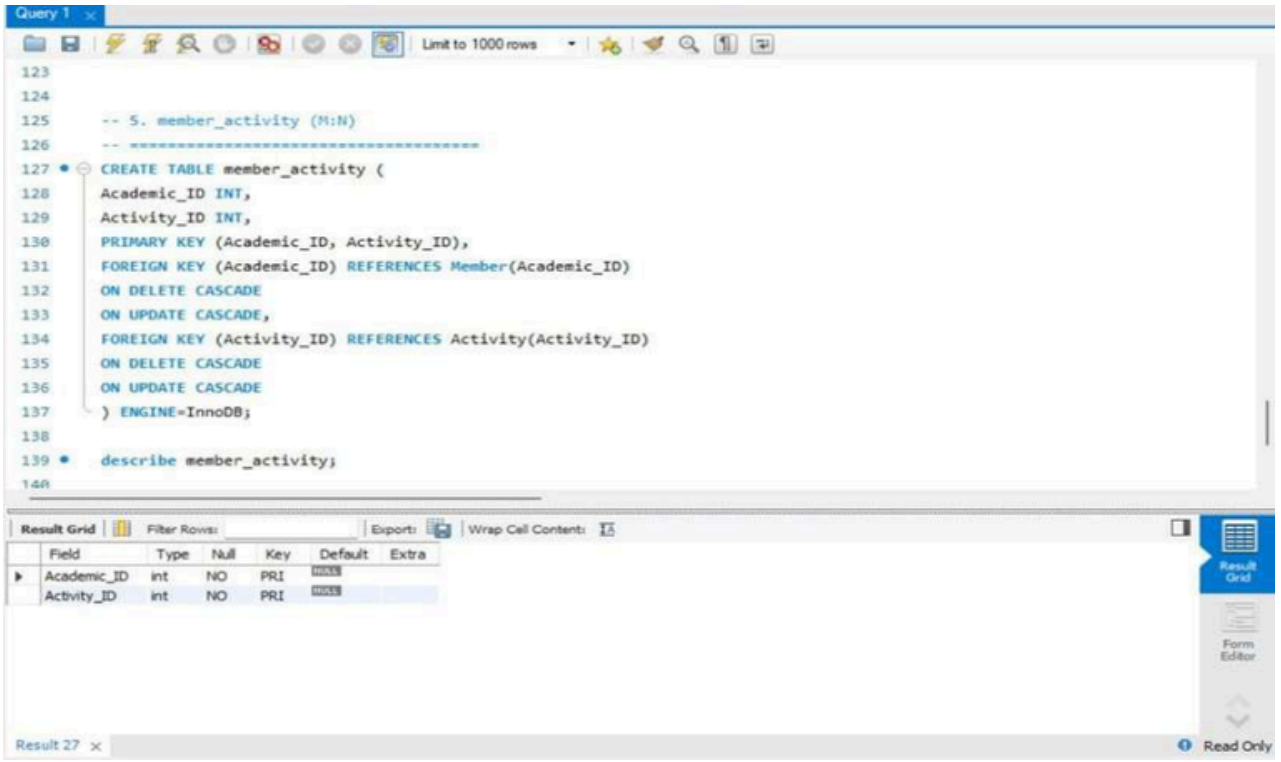


Image 13: Describe member_activity

Select from member_activity

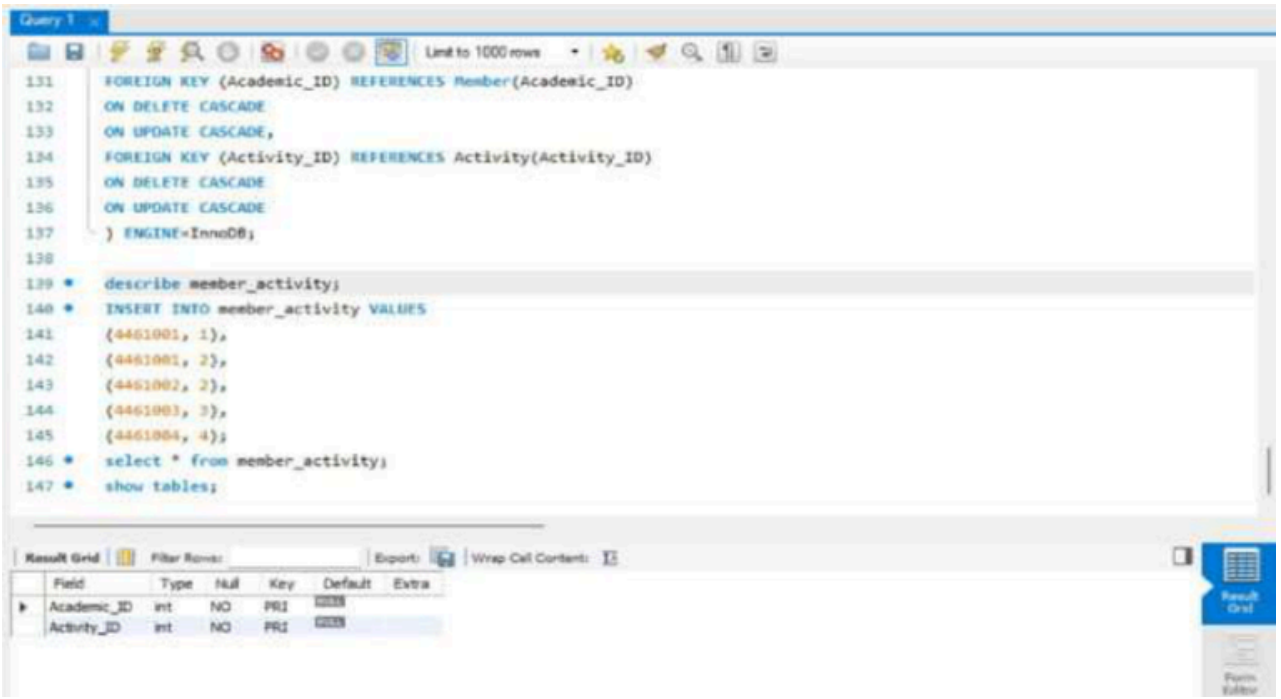


Image 14: Select from member_activity

Section 3

Display Member Names and Emails

```
150
151
152     -- 3
153     -- ------------------------------------
154 •   USE computing_club;
155
156 •   SELECT Fname, Email
157     FROM Member;
158
159
160
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Fname	Email
Sara	sara1@uqu.edu.sa
Jana	jana2@uqu.edu.sa
Jawana	jawana3@uqu.edu.sa
Ghala	ghala4@uqu.edu.sa
Layan	layan5@uqu.edu.sa

Member 9 | Read Only

Image 15: Display member names and emails

Count Members by College

Query 1

```
162
163
164
165
166 •   SELECT College, COUNT(*) AS Number_of_Members
167     FROM Member
168     GROUP BY College;
169
170
171
172
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

College	Number_of_Members
Computer Science	5

Result 10 | Read Only

Image 16: Count the number of members in each college

Display Workshop Activities

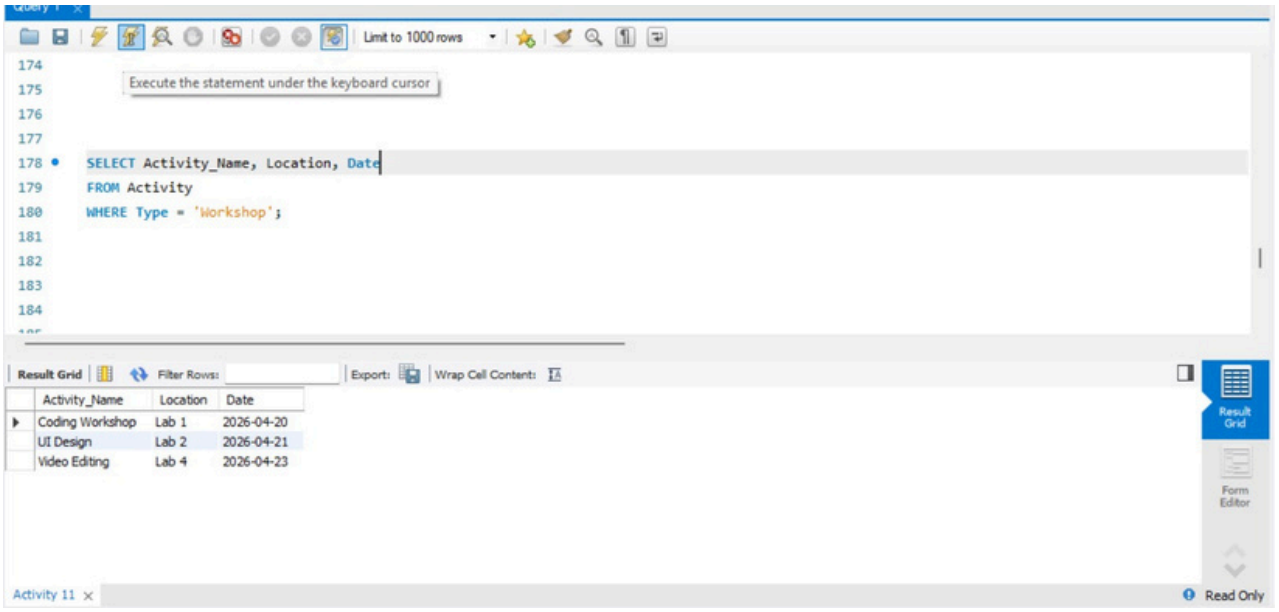


Image 17: Display workshop activities

Count Total Activities

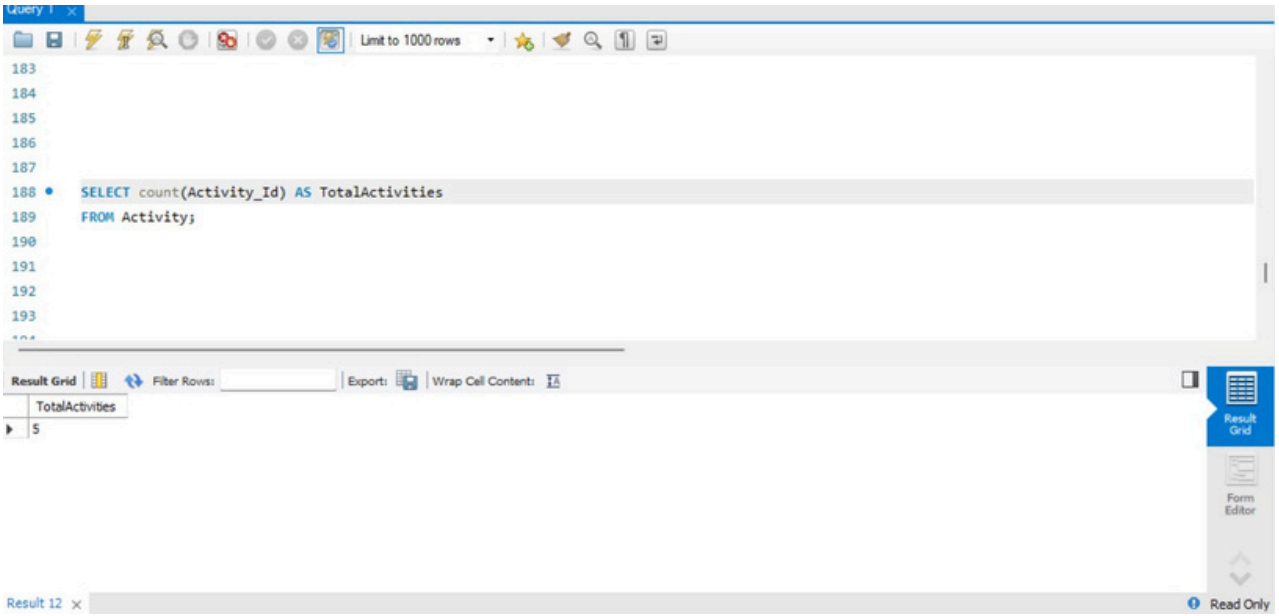


Image 18: Count total activities

Join Activity and Community Tables

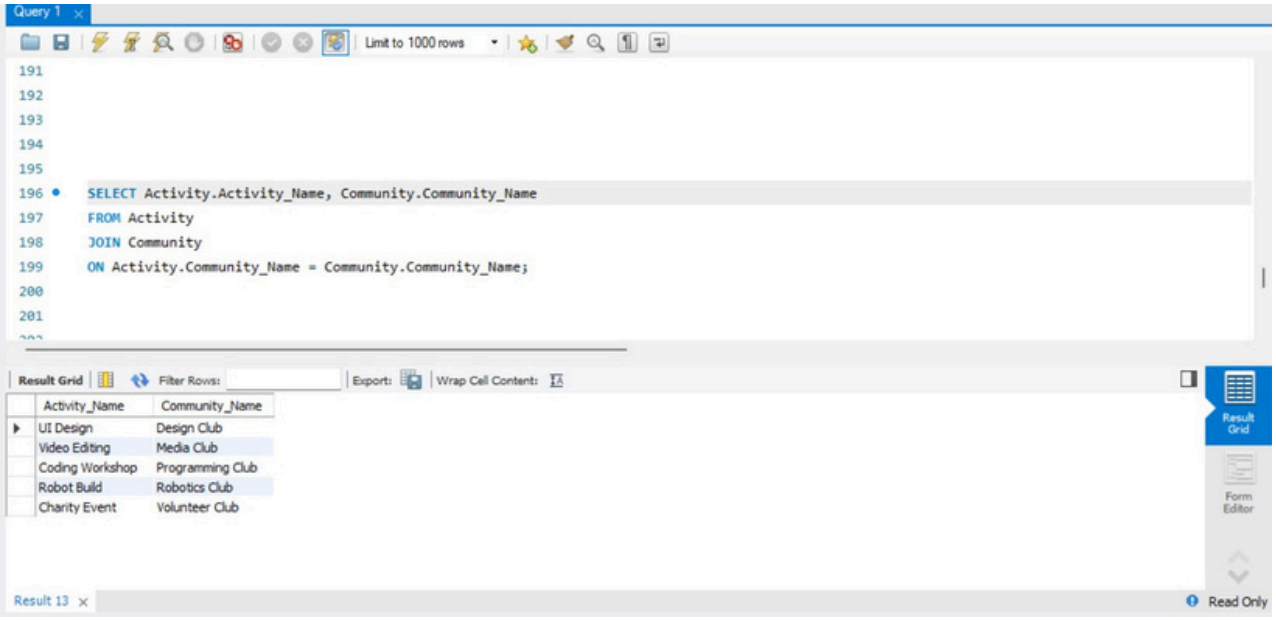


Image 19: Join between Activity and Community tables

Display Activities for Design Club Using Subquery

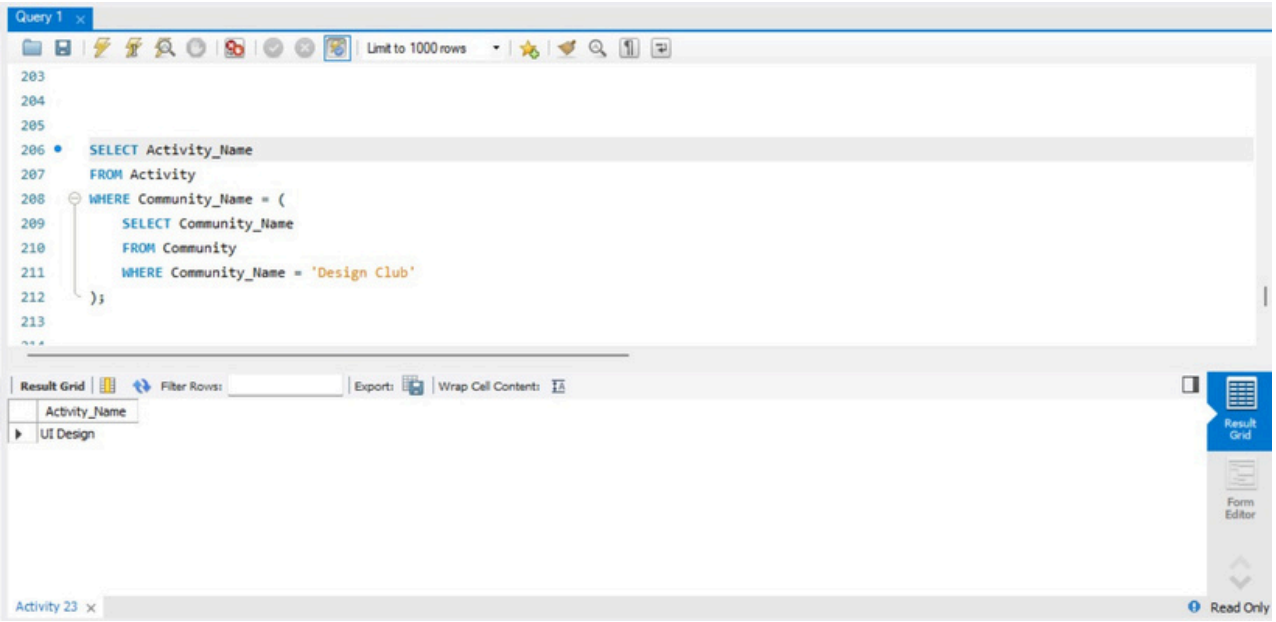


Image 20: Display Activities for Design Club Using Subquery

Display Activity_Report View

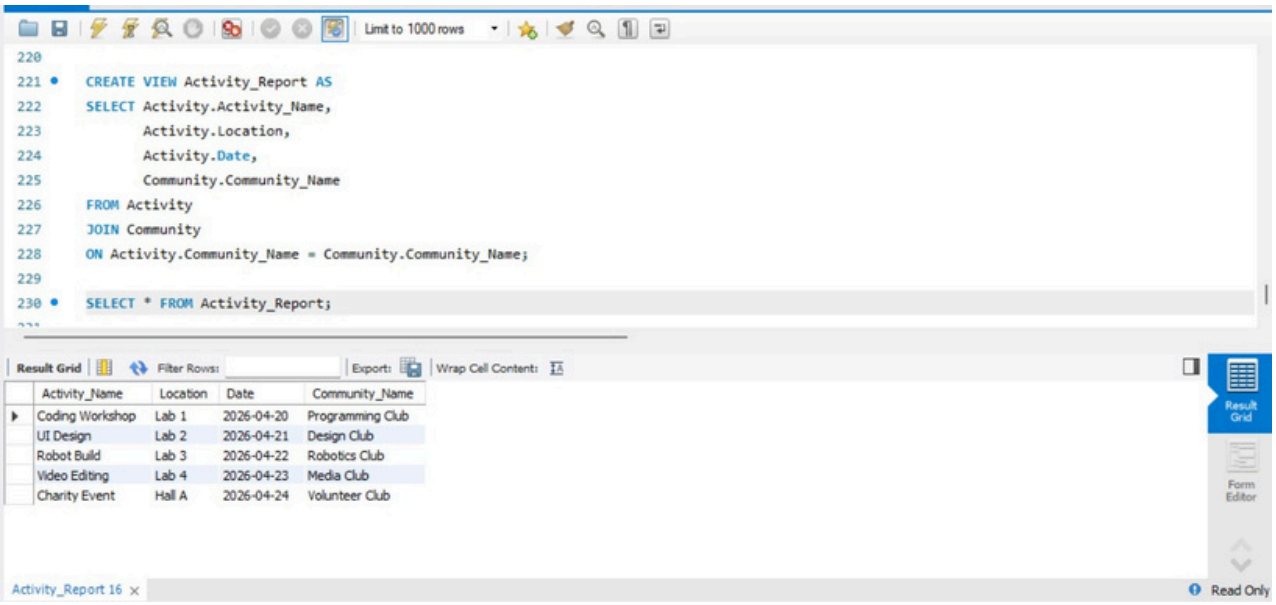


Image 21: Display data from Activity_Report view